

Tanya **Strydom**

Postdoctoral Researcher

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Research Profile

My research develops a theory of ecological inference for complex systems. I investigate how ecological representations capture, distort, and constrain our understanding of biodiversity. By combining ecological theory, network science, machine learning, and complexity science, I study how interactions emerge across scales, the propagation of uncertainty through ecological analyses, and what can ultimately be inferred and predicted about ecological systems from incomplete observations. My work aims to transform ecological networks from descriptive summaries into rigorous, predictive tools for biodiversity science and decision-making.

Professional Experience

Postdoctoral Researcher — *Sheffield, UK*

University of Sheffield — *2024 - Present*

Education

Doctor of Philosophy in Biological Sciences — *Montréal, Canada*

Université de Montréal — *2020-24*

Master of Science in Ecology and Biodiversity — *Stockholm, Sweden*

Stockholms Universitet — *2018-20*

Bachelor of Science (Honours) in Plant Sciences — *Pretoria, South Africa*

University of Pretoria — *2017*

Bachelor of Science in Ecology — *Pretoria, South Africa*

University of Pretoria — *2014-16*

Funding and Awards

Robert May Prize

Awarded by: British Ecological Society *2022*

Qualified for the UP Postgraduate Masters Research Bursary

Awarded by: University of Pretoria *2018*

Awarded the 3rd year Undergraduate Mentorship Bursary

Awarded by: University of Pretoria *2016*

Research Outputs

Publications

* Indicates co-lead author

14. Trepel, J., Buitenwerf, R., Ferraro, K., le Roux, E., Lundgren, E., Maitner, B. S., **Strydom, T.**, Svenning, J.-C., & Kerr, M. (2026). Reevaluating introduced herbivores in conservation. *Conservation Biology*. https://doi.org/10.1111/cobi.70328
13. Banville, F., **Strydom, T.**, Blyth, P. S. A., Brimacombe, C., Catchen, M., Dansereau, G., Gravel, D., Higino, G., Malpas, T., Mayall, H., Norman, K., & Poisot, T. (2025). Deciphering probabilistic species interaction networks. *Ecology Letters*, *28*(6). https://doi.org/10.1111/ele.70161
12. Higino, G. T., Anujan, K., Boakye, M., Degano, M. E., Forero-Muñoz, N.-R., & **Strydom, T.** (2025). Designing a collective prototype of future (sub)tropical science. *FACETS*, *10*. https://doi.org/10.1139/facets-2024-0113
11. Johnson, T. F., Simmons, B. I., Millard, J., **Strydom, T.**, Danet, A. H., Sweeny, A. R., & Evans, L. C. (2024). Pressure to publish introduces LLM risks. *Methods in Ecology and Evolution*, *15*(10). https://doi.org/10.1111/2041-210X.14397
10. Christiansen, D. M., **Strydom, T.**, Greiser, C., McClory, R., Ehrlén, J., & Hylander, K. (2023). Effects of past and present microclimates on northern and southern plant species in a managed forest landscape. *Journal of Vegetation Science*. https://doi.org/10.1111/jvs.13197
9. Maitner, B. S., Halbritter, A. H., Telford, R. J., **Strydom, T.**, Chacón-Labella, J., Henderson, A. N., Lamanna, C., Sloat, L. L., Kerkhoff, A. J., Messier, J., Rasmussen, N. L., Pomati, F., Merz, E., Vandvik, V., & Enquist, B. J. (2023). Bootstrapping outperforms community-weighted approaches for estimating the shapes of phenotypic distributions. *Methods in Ecology and Evolution*. https://doi.org/10.1111/2041-210X.14160
8. **Strydom, T.**, Bouskila, S., Banville, F., Barros, C., Caron, D., Farrell, M. J., Fortin, M.-J., Hemming, V., Mercier, B., Pollock, L. J., Runghen, R., Dalla Riva, G. V., & Poisot, T. (2023). Graph embedding and transfer learning can help predict potential species interaction networks despite data limitations. *Methods in Ecology and Evolution*, *14*(12). https://doi.org/10.1111/2041-210X.14228
7. **Strydom, T.**, & Poisot, T. (2023). SpatialBoundaries.jl: Edge detection using spatial wombling. *Ecography*. https://doi.org/10.1111/ecog.06609
6. Raath-Krüger, M. J., Schöb, C., McGeoch, M. A., Burger, D. A., **Strydom, T.**, & le Roux, P. C. (2022). Long-term spatially-replicated data show no cost to a benefactor species in a facilitative plant-plant interaction. *Oikos*. https://doi.org/10.1111/oik.09617
5. **Strydom*, T.**, Bouskila*, S., Banville, F., Barros, C., Caron, D., Farrell, M. J., Fortin, M.-J., Hemming, V., Mercier, B., Pollock, L. J., Runghen, R., Dalla Riva, G. V., & Poisot, T. (2022). Food web reconstruction through phylogenetic transfer of low-rank network representation. *Methods in Ecology and Evolution*. https://doi.org/10.1111/2041-210X.13835
4. Chacón-Labella, J., Boakye, M., Enquist, B. J., Farfan-Rios, W., Gya, R., Halbritter, A. H., Middleton, S. L., von Oppen, J., Pastor-Ploskonka, S., **Strydom, T.**, Vandvik, V., & Geange, S. R. (2021). From a crisis to an opportunity: Eight insights for doing science in the COVID-19 era and beyond. *Ecology and Evolution: Academic Practice in Ecology and Evolution*, *11*(8), 3588–3596. https://doi.org/10.1002/ece3.7026
3. Geange*, S. R., von Oppen*, J., **Strydom*, T.**, Boakye, M., Gauthier, T.-L. J., Gya, R., Halbritter, A. H., Jessup, L. H., Middleton, S. L., Navarro, J., Pierfederici, M. E., Chacón-Labella, J., Cotner, S., Farfan-Rios, W., Maitner, B. S., Michaletz, S. T., Telford, R. J., Enquist, B. J., & Vandvik, V. (2021). Next-generation field courses: Integrating Open Science and online learning. *Ecology and Evolution: Academic Practice in Ecology and Evolution*, *11*(8), 3577–3587. https://doi.org/10.1002/ece3.7009

2. **Strydom, T.**, Dalla Riva, G. V., & Poisot, T. (2021). SVD entropy reveals the high complexity of ecological networks. *Frontiers in Ecology and Evolution*, *9*. https://doi.org/10.3389/fevo.2021.623141
1. **Strydom*, T.**, Catchen*, M. D., Banville, F., Caron, D., Dansereau, G., Desjardins-Proulx, P., Forero-Muñoz, N. R., Higino, G., Mercier, B., Gonzalez, A., Gravel, D., Pollock, L. J., & Poisot, T. (2021). A roadmap toward predicting species interaction networks (across space and time). *Philosophical Transactions of the Royal Society B*, *376*(20210063). https://doi.org/10.1098/rstb.2021.0063

Preprints

5. Catchen, M. D., Banville, F., Blyth, P., Brimacombe, C., Dansereau, G., Higino, G., Malpas, T., Mayall, H., Norman, K., **Strydom, T.**, Pollock, L. J., Gonzalez, A., & Poisot, T. (2026). Designing optimal biodiversity observation networks. *EcoEvoRxiv*. https://doi.org/10.32942/X22T28
4. Karapunar, B., **Strydom, T.**, Beckermann, A. P., Ridgewell, A., Little, C., Pimiento, C., Dunne, J. A., Hull, P., Wignall, P., & Dunhill, A. M. (2026). No global collapse of food webs across the permian-triassic mass extinction. *BioRxiv*. https://doi.org/10.64898/2026.02.24.707709
3. Karapunar, B., **Strydom, T.**, Beckermann, A. P., Ridgewell, A., Little, C., Pimiento, C., Dunne, J. A., Hull, P., Wignall, P., & Dunhill, A. M. (2026). Temporal resolution reshapes dynamics of inferred community structure and extinction selectivity across the permian-triassic mass extinction. *BioRxiv*. https://doi.org/10.64898/2026.06.22.733099
2. **Strydom, T.**, Dunhill, A. M., Dunne, J. A., Poisot, T., & Beckerman, A. P. (2026). From metawebs to realised webs: A framework for ecological network representation under global change. *EcoEvoRxiv*. https://doi.org/10.32942/X2JW8K
1. **Strydom, T.**, Karapunar, B., Dunne, J. A., Hull, P., Little, C. T. S., Pimiento, C., Ridgewell, A., Beckermann, A. P., & Dunhill, A. M. (2026). Are we mapping ecosystems or models? Framework choices dominate food web topology and extinction inferences. *EcoEvoRxiv*. https://doi.org/10.32942/X2XQ1M

Archived datasets

2. Halbritter, A. H., Vandvik, V., Cotner, S., Farfan-Rios, W., Maitner, B. S., Michaletz, S. T., Menor, I. O., Telford, R. J., Ccahuana, A., Cruz, R., Bravo, J. S., Andrade, P. E. S., Bustamante, L. L. V., Castorena, M., Chacon-Labella, J., Christiansen, C. T., Duran, S. M., Egelkraut, D. D., Gya, R., … Enquist, B. J. (2024). Plant trait and vegetation data along a 1314 m elevation gradient with fire history in Puna grasslands, Perú. *Scientific Data*, *11*(225). https://doi.org/10.1038/s41597-024-02980-3
1. Kattge, J., Boenisch, G., Diaz, S., Lavorel, S., Prentice, C., Leadley, P., Wirth, C., & the TRY Consortium. (2020). TRY plant trait database–enhanced coverage and open access. *Global Change Biology*, *26*(1), 119–188. https://doi.org/10.1111/gcb.14904

Collaborative Projects

Metaweb2

PIs: Merin Reji Chacko — *2026*

Black Holes and Revelations: Identifying Priority Sampling Locations for Local Food Webs in Canada

PIs: F. Banville, M. Catchen, G. Dansereau, and **T. Strydom** — *2022*

Merging Statistical Theory and Analyses at the Interface of Microbial and 'Macrobial' Ecology

PIs: M. Liebold, P. Peres-Neto, and E. Thebault — *2022*

Canadian metaweb construction working group

PIs: T. Poisot and L.J. Pollock — *2021*

Network prediction synthesis working group

PIs: T. Poisot and L.J. Pollock — *2020*

Plant Functional Trait Course 5 in Peru

PIs: V. Vandvik and B.J. Enquist — *2020*

Presentations

Workshops and Organised Sessions

Space Oddity: Thinking About Ecological Networks Across Space — *ESA/CSEE Meeting*
Francis Banville, Gabriel Dansereau, **Tanya Strydom** — *Aug., 2022*

Designing a collective prototype of future tropical and subtropical science — *ATBC Annual Meeting*
Gracielle Higino, Mickey Boakye, Norma Forero, **Tanya Strydom** — *Jul., 2021*

Invited talks and seminars

Take a Byte: How the (En)Codification of Food Webs Impacts Inference — *Theoretical Ecology Seminar Series*
Tanya Strydom — *Oct., 2026*

How do you Reconstruct a Food Web that Died 200 million Years Ago? — *Yorkshire Geological Society*
Tanya Strydom — *Oct., 2026*

The Risks and Rewards of Large Language Models in (Open) Scientific Research — *SORTEE*
Tanya Strydom — *Oct., 2025*

What's [complexity] got to do with it? — *NetSci 2023*
Tanya Strydom — *Jul., 2023*

Making something out of nothing at all: Transfer learning for network prediction — *ML4MS mini-conference*
Tanya Strydom — *Apr., 2022*

Taking FAIR and open science to the field: The evolution of the PFTC field course — *Living Norway Colloquium*
Tanya Strydom alongside Aud H. Halbritter, 109 PFTC Participants — *Oct., 2020*

Talks

It's okay to be different: Understanding how and why network construction matters — *4th CPEG Symposium*
Tanya Strydom — *July, 2025*

Exploring the complexity of ecological networks using SVD entropy — *11th Annual QCBS Symposium*
Tanya Strydom, Giulio V. Dalla Riva and Timothée Poisot — *Dec., 2020*

Short Presentations and Posters

The Nuances of Food Webs: An Overview of Definitions, Scales, and Mechanisms — *BES Annual Meeting*
Tanya Strydom, Jennifer Dunne, Timothée Poisot and Andrew P. Beckerman — *Dec., 2024*

Food web reconstruction using transfer learning — *12th Annual QCBS Symposium*
Tanya Strydom, Salomé Bouskila and Timothée Poisot — *Dec., 2021*

Reconstructing food webs using transfer learning — *CSEE-SCEE Annual Meeting*
Tanya Strydom, Salomé Bouskila and Timothée Poisot — *Aug., 2021*

Teaching

Invited Lectures

Species Interaction Networks: A Macro-level Overview — *Univerity of South Florida*
Community Ecology — *2025 - Present*

Science Communication & Outreach

Scientific outreach

Cartoonist

Fortnightly cartoonist for Ecology for the Masses blog *2020 - 2022*

Skype a Scientist

Q&A sessions with school groups *2024 - Present*

Letters to a PreScientist

Pen pal programme with high school students *2025*

Popular Articles

3. Cooper, N., Hsing, P.-Y., Almarzouq, B., Baldauf, S., Chatterjee, N., Plomp, E., **Strydom, T.**, Takola, E., & Zagrodzka, Z. (2025). *Guide to reproducible code*. `https://doi.org/10.5281/zenodo.17855982`
2. von Oppen, J., Gya, R., Geange, S., **Strydom, T.**, Middleton, S., & Maitner, B. S. (2021). Next generation field courses: Enhancing ECR development through open science and online learning. *Ecology for the Masses*.
1. Cotner, S., Enquist, B. J., Chacon, J., Maitner, B. S., Farfan-Rios, W., Michaletz, S., Garen, J., Gauthier, T.-L. J., Vandvik, V., Gya, R., Halbritter, A. H., Hošková, K., Pierfederici, M. E., Quinteros-Casaverde, N. L., Diaz, E. S., Jessup, L. H., **Strydom, T.**, & von Oppen, J. (2020). International scientists need better support during global emergencies. *Times Higher Education*.